

MISSION STATEMENT

S-ART (STUDENT ASTRONOMICAL RADIO TELESCOPE)

The **S-ART mission** consists of the development and operation of a **scientific-educational CubeSat** designed to observe the Sun from low Earth orbit. Its primary objective is to **detect and characterise electromagnetic emissions associated with dynamic solar phenomena**, particularly **coronal mass ejections (CMEs)** and low- and high-frequency radio bursts. Using a **dual-antenna payload**, S-ART aims to deliver complementary observations to those obtained by major solar missions, thereby improving the understanding of coronal physics, CME generation mechanisms, and their potential predictability.

S-ART is conceived as a **collaborative university mission** with a strong educational component, exposing students to the full lifecycle of space engineering through processes inspired by **ECSS standards**. The project integrates teams covering systems engineering, communications, payload development, AOCS, power, and operations.

The mission will operate in a **LEO orbit** and will have a lifetime shorter than five years to comply with deorbiting requirements. The spacecraft will be designed around a **robust and service-oriented platform**, including **redundant mechanisms for power distribution, attitude control, and data transmission** to ensure mission continuity and increase operational reliability. Additional **fault-management strategies** and **safe-mode capabilities** will be implemented to maximise data return throughout the mission.

S-ART aims to **generate valuable scientific data**, address observational gaps in specific radio bands, and demonstrate the capacity of the academic community to develop **reliable, relevant, and cost-effective space instrumentation**.